ME 5309

Mid-Semester Examination – Grading Rubric for Teaching Assistants

Part A – How to grade

A.1 Best three of four. Read and mark **every** question the candidate has attempted, including any fourth attempt, unless the candidate has struck it out. Enter all four scores in the tabulation sheet, then sum the **highest three**. Do not decide in advance which three to mark. If a candidate has struck out an attempt, do not mark it, even if it looks better than the others.

A.2 Mark granularity. Every nominal split in this rubric is a **whole mark**. Halves appear only where a common-error note explicitly awards one. Do not use quarter marks. Each question totals exactly 10.

A.3 Method before answer. Instruction 5 on the paper states that a final numerical answer without the formula written earns no credit. Apply that symmetrically:

- A **correct final answer with no formula written scores zero** for that part. This is an open-notes examination; an unsupported number is not evidence of understanding.
- A **correct formula with correct substitution but an arithmetic slip** retains the formula marks and loses only the answer mark. Mark the slip "AE" in the margin.
- A correct method with a wrong formula taken from the notes scores the substitution credit only if the substitution is consistent with the formula written.

A.4 Carry-through (apply once per question). If an early part of a question produces a wrong number and a later part of the **same** question uses that number correctly, do **not** penalise the later part again. Mark the later part on method and consistency. Penalise the original error once only. Carry-through does **not** apply across questions.

Part B – Answer key at a glance

Part	Key result	Marks
Q1(a)	$f_x = 3200$ pixels; field of view $= 160$ mm; one pixel $= 0.125$ mm on the belt	4
Q1(b)	LSB $= 1.22$ mV, force resolution $= 0.122$ N	3
Q1(c)	$F = 0.8187, \quad G = 0.09063$	3
Q2(a)	$mg\ell \sin\theta_e = \tau_0$; equilibrium exists for $\lvert\tau_0\rvert \le 2.4525$ N·m	2
Q2(b)	$A = \left[\begin{smallmatrix} 0 & 1 \\ -61.3 & -2.5 \end{smallmatrix}\right], \quad B = \left[\begin{smallmatrix} 0 \\ 50 \end{smallmatrix}\right]$	2
Q2(c)	$K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$; error ≈ 0.049 rad $= 2.81^\circ$	4
Q2(d)	Error $= 0$; gravity stiffness removed from the loop (negligible change at $\theta_d = 90^\circ$)	2
Q3(a)	Quantisation amplified by $1/T$; noise amplification by differentiation	2
Q3(b)	$y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}$; DC gain $= 1$	2
Q3(c)	$s^2 + (\ell_1 + 2)s + (2\ell_1 + \ell_2)$; $\quad L = \left[\begin{smallmatrix} 18 \\ 64 \end{smallmatrix}\right]$	4
Q3(d)	Observer uses input and model, estimates unmeasured state without lag; filter safer under model uncertainty	2
Q4(a)	Poles $-2 \pm 2j$; $\quad N_{ff} = 8$	2
Q4(b)	$\theta_{ss} = 0.800$, error $= -20\%$; N_{ff} is outside the feedback path	2
Q4(c)	$s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$; $\quad k_1 = 18$, $k_2 = 6$, $K_I = 32$	4
Q4(d)	$\dot{\xi} = 0 \Rightarrow \theta = r$ for any b, c; windup, cured by anti-windup	2

Part C – Worked solutions and mark allocation

Question 1 [10 marks]

Q1(a) – Camera scaling: focal length in pixels and field of view [4 marks]

(i) Focal length in pixels. From the intrinsics slide, f_x is the focal length expressed in pixel units, $f_x = f/(\text{pixel width})$:

$$f_x = \frac{16 \text{ mm}}{5 \times 10^{-3} \text{ mm}} = 3200 \text{ pixels}$$

(The pixels are square, so $f_y = f_x = 3200$ as well.)

(ii) Field of view and millimetres per pixel. The sensor is $1280 \times 5\ \mu\text{m} = 6.4$ mm wide. Using FOV $\approx$ sensor size $\times\, d_o/f$:

$$\text{FOV} = 6.4 \text{ mm} \times \frac{400}{16} = 6.4 \times 25 = 160 \text{ mm} \qquad \Rightarrow \qquad \text{one pixel} = \frac{160 \text{ mm}}{1280} = 0.125 \text{ mm on the belt}$$

Second valid route, equally acceptable. From the pinhole equation $u = f_x X/Z + c_x$, a displacement of one pixel corresponds to $\Delta X = Z/f_x = 400/3200 = 0.125$ mm, and the field of view follows as $1280 \times 0.125 = 160$ mm. A candidate who works in either direction earns full credit. Treat the standoff d_o and the depth Z as the same quantity here.

Credit	Marks
Correct $f_x = 3200$ pixels, from $f/(\text{pixel width})$	2
Correct field of view 160 mm	1
Correct 0.125 mm per pixel on the belt (by either route above)	1

Q1(b) – ADC force resolution [3 marks]

$$\text{LSB} = \frac{V_{FS}}{2^N} = \frac{5}{2^{12}} = \frac{5}{4096} = 1.2207 \text{ mV} \qquad \Rightarrow \qquad \Delta F = \frac{\text{LSB}}{\text{sensitivity}} = \frac{1.2207 \text{ mV}}{10 \text{ mV/N}} = 0.1221 \text{ N}$$

Credit	Marks
Correct LSB = 1.22 mV, from a written $V_{FS}/2^N$	2
Divides by the sensitivity to get 0.122 N	1

Common errors

- Uses $2^{12} - 1 = 4095 \rightarrow 1.2210$ mV, 0.1221 N. **Accept in full**; the distinction is not examinable here.
- Multiplies by the sensitivity instead of dividing. Award the 2 LSB marks only.
- Uses a range of 10 V or ± 5 V. Flag; award the LSB marks only if the arithmetic matches the stated range.
- Writes 1.22 mV with no formula above it. award 0 for the LSB marks, and the division mark only if the division itself is shown.

Q1(c) – Exact ZOH discretisation [3 marks]

For $\dot{x} = Ax + Bu$ held constant over a sampling interval T,

$$F = e^{AT}, \qquad G = \left(\int_0^T e^{A\tau}\, d\tau \right) B.$$

Here $A = -2$, $B = 1$, $T = 0.1$:

$$F = e^{-2(0.1)} = e^{-0.2} = 0.81873 \qquad G = \int_0^{0.1} e^{-2\tau}\, d\tau = \frac{1 - e^{-0.2}}{2} = 0.090635$$

Equivalently, using $G = A^{-1}(F - I)B = \dfrac{0.81873 - 1}{-2} = 0.090635$.

Credit	Marks
States both $F = e^{AT}$ and $G = \left(\int_0^T e^{A\tau} d\tau \right) B$ (or the $A^{-1}(F-I)B$ form)	1
Correct $F = 0.8187$	1
Correct $G = 0.0906$	1

Common errors

- Gives only F and leaves G blank. Award **2** (formulae + F).
- Uses forward Euler, $F = 1 + AT = 0.8$, $G = T = 0.1$. The question says *exact*, and the paper asks for the formulae to be stated first. Award **1** for a correct and clearly labelled Euler discretisation, no more.
- Uses Tustin. Same treatment as Euler: **1** maximum.
- Writes $G = A^{-1}(e^{AT} - I)B$ but inverts the sign $\rightarrow -0.0906$. Award the formula mark and the F mark. *Total 2.*
- Correct method but reports $F = e^{-0.2}$ unevaluated and G unevaluated. Award **1.5**: the formula mark in full and half the F mark for the correct exponent. That is Euler by another name. **1** maximum.

Question 2 [10 marks]

Q2(a) – State-space model and equilibria [2 marks]

With $x_1 = \theta$, $x_2 = \dot{\theta}$:

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = \frac{1}{J}(\tau - b x_2 - mg\ell \sin x_1)$$

At equilibrium $\dot{x}_1 = \dot{x}_2 = 0$, so $x_2 = 0$ and

$$mg\ell \sin\theta_e = \tau_0, \qquad mg\ell = 1 \times 9.81 \times 0.25 = 2.4525 \text{ N·m}$$

Since $|\sin\theta_e| \leq 1$, an equilibrium exists only for $|\tau_0| \leq 2.4525$ N·m.

Credit	Marks
Correct two-state model with both equations	1
Equilibrium condition $mg\ell \sin\theta_e = \tau_0$ **and** the existence range $\lvert\tau_0\rvert \leq 2.4525$ N·m	1

Common errors

- Correct equilibrium equation but no existence range. Award 0 for the second mark; the range is the point of the part.

- Gives the range as $\tau_0 \leq 2.4525$ without the modulus. Accept in full if the symmetry is mentioned anywhere; otherwise award 0.5.
- Gives the range symbolically as $|\tau_0| \leq mg\ell$ without evaluating. Award 0.5.
- Forgets that there are **two** equilibria for each admissible τ_0. Not required; do not penalise, but note it if the candidate claims uniqueness.

Q2(b) – Linearisation **[2 marks]**

$$A = \begin{bmatrix} 0 & 1 \\ -\dfrac{mg\ell\cos\theta_e}{J} & -\dfrac{b}{J} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ \dfrac{1}{J} \end{bmatrix}$$

At $\theta_e = 60^\circ$, $\cos 60^\circ = 0.5$:

$$\frac{mg\ell\cos\theta_e}{J} = \frac{2.4525 \times 0.5}{0.02} = 61.31, \qquad \frac{b}{J} = \frac{0.05}{0.02} = 2.5 \qquad \Rightarrow \qquad A = \begin{bmatrix} 0 & 1 \\ -61.3 & -2.5 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 50 \end{bmatrix}$$

Credit	Marks
Correct symbolic A and B, with the $\cos\theta_e$ term and the correct signs	1
Correct numerical entries -61.3, -2.5, 50	1

Common errors

- Uses $\sin\theta_e$ instead of $\cos\theta_e$ in $A_{21} \to -106.2$. Lose the symbolic mark; award the numerical mark only if consistent with their own expression. *Maximum 1.*
- Sign error on A_{21}. Lose the symbolic mark. *Maximum 1.*
- Correct A but $B = \begin{bmatrix} 0 & 1 \end{bmatrix}^\mathsf{T}$, forgetting the $1/J$. Lose the numerical mark. *Maximum 1.*
- Measures $\theta_e = 60^\circ$ from the horizontal, giving $\cos 30^\circ = 0.866$ and $A_{21} = -106.2$. The paper defines the convention. Lose the symbolic mark; flag if widespread.

Q2(c) – Steady-state droop under pure PD **[4 marks]**

Substituting the PD law into the plant:

$$J\ddot{\theta} + b\dot{\theta} + mg\ell\sin\theta = K_p(\theta_d - \theta) - K_d\dot{\theta}$$

At a steady state, $\ddot{\theta} = \dot{\theta} = 0$:

$$\boxed{K_p\,(\theta_d - \theta_{ss}) = mg\ell\sin\theta_{ss}}$$

$\theta_{ss} = \theta_d$ cannot satisfy this, because the left side would be zero while the right side is $mg\ell\sin 90^\circ = 2.4525 \neq 0$. There must therefore be a non-zero error: the proportional term has to generate the torque that holds the load against gravity, and it can only do so by sitting at an angle away from the command.

Writing $e = \theta_d - \theta_{ss}$ and linearising $\sin\theta_{ss} \approx \sin\theta_d - e\cos\theta_d$ with $\theta_d = 90^\circ$ (so $\cos\theta_d = 0$):

$$e \approx \frac{mg\ell\sin\theta_d}{K_p + mg\ell\cos\theta_d} = \frac{2.4525}{50} = 0.04905 \text{ rad} = 2.81^\circ$$

Solving the transcendental equation numerically gives $\theta_{ss} = 87.19^\circ$, that is $e = 2.807^\circ = 0.04899$ rad – the estimate above is accurate to better than 0.2% here.

Credit	Marks
Shows that $\theta_{ss} = \theta_d$ is impossible, by substitution or by a torque-balance argument	1
Correct steady-state equation $K_p(\theta_d - \theta_{ss}) = mg\ell\sin\theta_{ss}$	1
Correct error value, 0.049 rad or 2.81°	2

Accept for the final two marks: any value in 2.7°–2.9°, or 0.047–0.050 rad, by any route – the linearised estimate, iteration on the transcendental equation, or graphical solution. Accept the answer expressed as $\theta_{ss} \approx 87.2°$. The paper asks for the error *in degrees*, so an answer left in radians attracts the single unit penalty of A.6 if no conversion is given.

Common errors

- States "PD gives steady-state error because there is no integral action" and stops, with no equation. Award 1 for the qualitative point; the remaining 3 require the equation and the number.
- Writes the steady-state equation correctly but then uses $\sin\theta_{ss} \approx \theta_{ss}$ (small-angle about zero, invalid at 90°). Award the first 2, lose the final 2.
- Computes $e = mg\ell/K_p$ directly without justifying the approximation. **Award in full** – the justification at $\theta_d = 90°$ is that $\cos\theta_d = 0$, and arriving at the right expression is sufficient evidence.
- Sets up the equation and solves it numerically but reports $\theta_{ss} = 87.2°$ without stating the error. Award 3; the question asks for the error.
- Mixes operating points, using $\cos 60°$ from part (b) in the denominator $\rightarrow e = 0.0472$ rad $= 2.70°$. Award the first 2 and **1** of the final 2, since the value happens to fall inside the tolerance band by coincidence rather than by method.

Q2(d) – Gravity compensation **[2 marks]**

The gravity term cancels exactly, leaving a **linear** closed loop:

$$J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$$

so $\theta_{ss} = \theta_d$ and the steady-state error is **zero**, for any θ_d.

On the poles, the compensated loop has

$$s^2 + \frac{b + K_d}{J}s + \frac{K_p}{J} = 0,$$

whereas the uncompensated loop, linearised about its own equilibrium, has

$$s^2 + \frac{b + K_d}{J}s + \frac{K_p + mg\ell\cos\theta_{ss}}{J} = 0.$$

The compensation therefore removes the gravity stiffness $mg\ell\cos\theta$ from the loop, so in general the poles **do** change. At this particular command, $\theta_d = 90°$, the gravity stiffness is $mg\ell\cos 87.2° = 0.12$ N·m/rad against $K_p = 50$, so the numerical change is under 0.3% – negligible here, but it would be large near $\theta = 0$ or 180°.

Credit	Marks
States the steady-state error is zero, with the cancellation shown or clearly asserted	1
Correct, justified statement about the poles	1

For the pole mark, award 1 for any of the following:

- "The poles change, because the $mg\ell\cos\theta$ stiffness term is removed from A" – the general answer.
- "At $\theta_d = 90°$ the gravity stiffness is zero, so the poles are essentially unchanged *at this operating point*, though they would change elsewhere" – the sharper answer.
- Both together – the best answer. There are no bonus marks, but note it on the script; these candidates are worth identifying.

Do not award for an unjustified "yes" or "no", or for "the poles change because the plant is now linear" without identifying which coefficient moves.

Question 3 [10 marks]

Q3(a) – Why raw differencing fails [2 marks]

Two distinct mechanisms are required, one mark each:

1. **Quantisation amplification.** The encoder output is quantised to one count. Differencing gives $\hat{\dot{\theta}} = (\theta_k - \theta_{k-1})/T$, so a one-count uncertainty becomes a velocity error of $\Delta\theta/T$ – the quantisation step divided by a small number. For an encoder resolving 0.01° per count, one count of jitter at 1 kHz is $0.01^\circ \times 1000 = 10^\circ$/s of spurious velocity.
2. **Noise amplification by differentiation.** Differentiation has a gain rising with frequency, so any broadband sensor or electrical noise on the position signal is amplified most at exactly the frequencies where it carries no useful information.

Credit	Marks
First valid mechanism, correctly explained	1
Second, genuinely distinct mechanism, correctly explained	1

Common errors

- Gives only one reason, however well explained. *Maximum 1.*
- Gives "noise" twice in different words: quantisation noise and sensor noise are the two intended answers, but "it is noisy" and "noise gets amplified" count as one mechanism. Award 1.
- Names a mechanism correctly but with no explanation of why differencing causes it, for example the single word "quantisation". Award 0.5 for that mechanism.
- Discusses the effect of increasing f_s. This was asked in an earlier draft but **not** in the final paper. It earns no extra credit and costs nothing; ignore it.

Q3(b) – Tustin discretisation of the low-pass filter [2 marks]

Continuous filter $H(s) = \dfrac{1}{1+s/\omega_c}$ with $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin substitution $s \to \dfrac{2}{T}\dfrac{1-z^{-1}}{1+z^{-1}}$ with $T = 1$ ms. Writing $\kappa = \dfrac{\omega_c T}{2} = \dfrac{125.66}{2000} = 0.062832$:

$$H(z) = \frac{\kappa\left(1+z^{-1}\right)}{(1+\kappa)+(\kappa-1)z^{-1}}, \qquad b_0 = b_1 = \frac{\kappa}{1+\kappa} = 0.059117, \qquad a_1 = \frac{\kappa-1}{1+\kappa} = -0.881765$$

$$\therefore \quad y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}$$

DC gain check at $z = 1$, which is the steady-state ratio of output to a constant input:

$$\frac{b_0+b_1}{1+a_1} = \frac{0.118235}{0.118235} = 1.000$$

Credit	Marks
Correct difference equation with both coefficients to 3 s.f.	1
DC gain evaluated at $z = 1$ and found to be unity	1

Q3(c) – Observer design **[4 marks]**

$$A - LC = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} \ell_1 \\ \ell_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} -\ell_1 & 1 \\ -\ell_2 & -2 \end{bmatrix}$$

$$\det\left(sI - (A - LC)\right) = \det \begin{bmatrix} s + \ell_1 & -1 \\ \ell_2 & s + 2 \end{bmatrix} = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (\ell_1 + 2)\, s + (2\ell_1 + \ell_2)$$

Desired polynomial for a double pole at -10: $(s+10)^2 = s^2 + 20s + 100$. Matching coefficients:

$$\ell_1 + 2 = 20 \;\Rightarrow\; \ell_1 = 18, \qquad 2\ell_1 + \ell_2 = 100 \;\Rightarrow\; \ell_2 = 100 - 36 = 64, \qquad L = \begin{bmatrix} 18 \\ 64 \end{bmatrix}$$

Credit	Marks
Correct $A - LC$ formed, with LC as an outer product placed in the first column	1
Correct characteristic polynomial $s^2 + (\ell_1 + 2)s + (2\ell_1 + \ell_2)$	1
Correct desired polynomial $s^2 + 20s + 100$	1
Correct $\ell_1 = 18$ and $\ell_2 = 64$	1

Q3(d) – Observer versus filter **[2 marks]**

Observer advantage (any one, 1 mark):

- It uses the known control input u and the plant model, so it predicts rather than differentiates: within its design bandwidth it tracks a changing velocity with no steady-state lag, whereas a low-pass filter always trades lag against smoothing.
- It produces an estimate of a state that is not measured at all, not merely a smoothed version of a differentiated measurement.
- The estimate is the state of a stable dynamic system driven by the innovation, so quantisation enters through a bounded, filtered path rather than being divided by T.

Filter is the safer choice (any one, 1 mark):

- The filter is model-free. The observer's estimate is biased whenever A, B or the actuator gain are wrong, and that bias grows with model error; a filter degrades only as its own bandwidth choice is wrong.
- If plant parameters drift (temperature, load, wear) or the plant is substantially nonlinear away from the linearisation point, the filter keeps working while the observer's prediction term becomes actively misleading.
- A filter cannot be destabilised by a modelling sign error; a badly identified model can make an observer's estimate diverge in a way that is hard to diagnose from the position signal alone.

Credit	Marks
One valid observer advantage, correctly reasoned	1
One valid circumstance favouring the filter, specifically invoking model error or parameter uncertainty	1

Question 4 **[10 marks]**

Q4(a) – Nominal design **[2 marks]**

Substituting $u = N_{ff} r - k_1\theta - k_2\dot{\theta}$:

$$\ddot{\theta} + (a + bk_2)\,\dot{\theta} + (c + bk_1)\,\theta = bN_{ff}\,r$$

With $a = 2$, $b = 1$, $c = 6$, $k_1 = k_2 = 2$:

$$s^2 + 4s + 8 = 0 \quad \Longrightarrow \quad s = -2 \pm 2j$$

DC gain: setting all derivatives to zero, $\theta_{ss} = \dfrac{bN_{ff}}{c + bk_1}\,r = \dfrac{N_{ff}}{8}\,r$, so unity DC gain requires

$$N_{ff} = \frac{c + bk_1}{b} = 8.$$

Credit	Marks
Correct closed-loop poles $-2 \pm 2j$	1
Correct $N_{ff} = 8$, obtained from a stated DC-gain condition	1

Q4(b) – Sensitivity of the feedforward design **[2 marks]**

With $b = 0.75$ and the gains unchanged:

$$\theta_{ss} = \frac{bN_{ff}}{c + bk_1} = \frac{0.75 \times 8}{6 + 0.75 \times 2} = \frac{6}{7.5} = 0.800 \qquad \Longrightarrow \qquad \text{error} = -20\%$$

N_{ff} is a **feedforward** gain: it sits on the reference path, upstream of the plant and outside the feedback loop. The feedback terms $k_1\theta$ and $k_2\dot{\theta}$ act on the measured state, so they can suppress disturbances and shape the transient, but at DC they have no information about whether the reference-to-output calibration is right. There is no integrator and no error signal driving a correction, so the steady-state gain inherits the actuator gain error directly.

Credit	Marks
Correct $\theta_{ss} = 0.800$ and error -20%	1
Explanation identifying N_{ff} as being outside the feedback path, hence uncorrected	1

Q4(c) – Integral action design **[4 marks]**

Take the augmented state $\mathbf{x} = \begin{bmatrix}\theta & \dot{\theta} & \xi\end{bmatrix}^{\mathsf{T}}$ with $\dot{\xi} = r - \theta$. Substituting $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$ into the plant:

$$\ddot{\theta} = -\,(c + bk_1)\,\theta - (a + bk_2)\,\dot{\theta} + bK_I\,\xi$$

$$A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ -(c + bk_1) & -(a + bk_2) & bK_I \\ -1 & 0 & 0 \end{bmatrix} \qquad \Longrightarrow \qquad \det(sI - A_{cl}) = s^3 + (a + bk_2)\,s^2 + (c + bk_1)\,s + bK_I$$

Desired poles -4 and $-2 \pm 2j$:

$$(s + 4)\left(s^2 + 4s + 8\right) = s^3 + 8s^2 + 24s + 32$$

Matching coefficients with $a = 2$, $b = 1$, $c = 6$:

$$a + bk_2 = 8 \;\Rightarrow\; k_2 = 6, \qquad c + bk_1 = 24 \;\Rightarrow\; k_1 = 18, \qquad bK_I = 32 \;\Rightarrow\; K_I = 32$$

Credit	Marks
Augmented third-order model formed correctly, with the integral state included	1
Correct symbolic characteristic polynomial $s^3+(a+bk_2)s^2+(c+bk_1)s+bK_I$	1
Correct desired polynomial $s^3+8s^2+24s+32$	1
All three gains correct: $k_1=18$, $k_2=6$, $K_I=32$	1

Note on sign conventions – important. The paper's convention ($\dot{\xi}=r-\theta$ with $+K_I\xi$ in the control) and the equally common alternative ($\dot{\xi}=\theta-r$ with $-K_I\xi$) give the **identical** characteristic polynomial and the identical gains. Accept either without comment. A candidate who uses $\dot{\xi}=r-\theta$ together with $-K_I\xi$ will find $K_I=-32$; **award in full** if they state the convention they adopted and their algebra is consistent, since the physical controller is the same.

Q4(d) – Robustness of integral action, and windup [2 marks]

Zero steady-state error for any b, c. At any steady state all derivatives vanish, including $\dot{\xi}$. But $\dot{\xi}=r-\theta$, so $\dot{\xi}=0$ forces

$$\theta=r$$

exactly, with no reference to a, b or c. The integrator state ξ simply settles at whatever value supplies the required constant control effort, $\xi_{ss}=\dfrac{(c+bk_1)\,r}{bK_I}$; its *value* depends on the parameters, but the *error it integrates* must be zero or it would not have stopped moving. The only requirements are that such a steady state be reached, that is, that the closed loop be stable, and that $b\neq 0$. This is the internal model principle: the controller contains a model of the constant reference, $1/s$.

Practical failure mode and remedy (any one):

- **Integrator windup.** When the actuator saturates, the plant can no longer follow, but the integrator continues to accumulate error. ξ grows far beyond the value needed, and when the error finally reverses the controller must unwind that accumulation, producing large overshoot and a long recovery. **Remedy:** anti-windup – stop or clamp integration while the actuator is saturated (conditional integration), or use back-calculation, feeding the difference between commanded and delivered control back into the integrator.
- **Reduced stability margin.** Integral action adds a pole at the origin and phase lag, so an aggressive K_I reduces the gain and phase margins. **Remedy:** place the integral pole well inside the dominant closed-loop poles, or add a derivative/feedforward path.
- **Integration of sensor bias.** A constant sensor offset is integrated indefinitely, driving a steady actuator effort against a non-existent error. **Remedy:** sensor calibration and zeroing, or a leaky integrator.

Credit	Marks
Argument that $\dot{\xi}=0\Rightarrow\theta=r$ regardless of b and c	1
One named failure mode **with** its remedy	1

Google Sheet

Please make a shared google sheet for us only. Enter all attempted question scores, each out of 10. Sum the best three, out of 30 in the following form.

Roll No.	Q1	Q2	Q3	Q4	Best three	Total /30	Referred?